Reality Gap Modeling & RL Motion Optimization

(simulated + physical)

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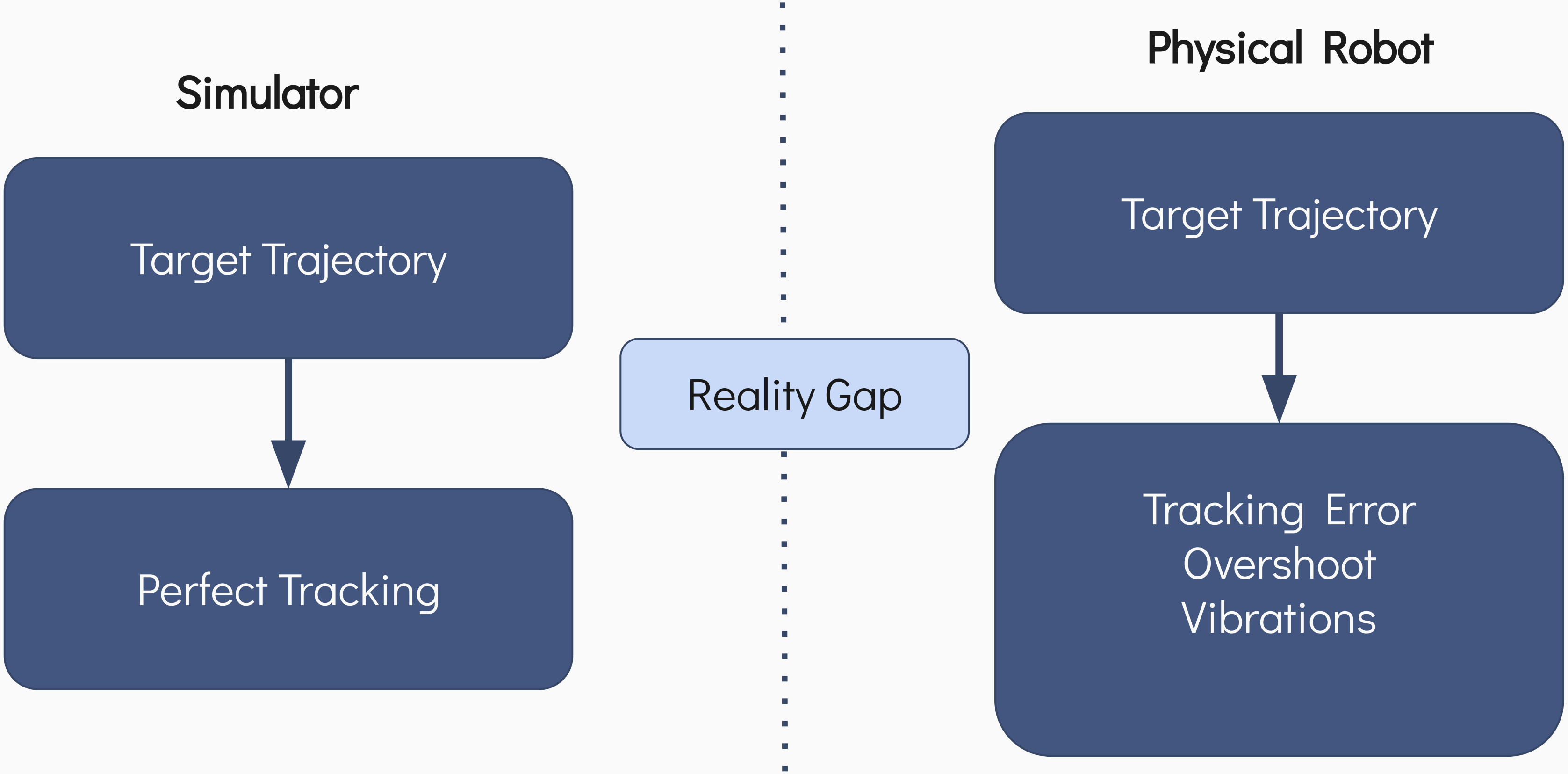
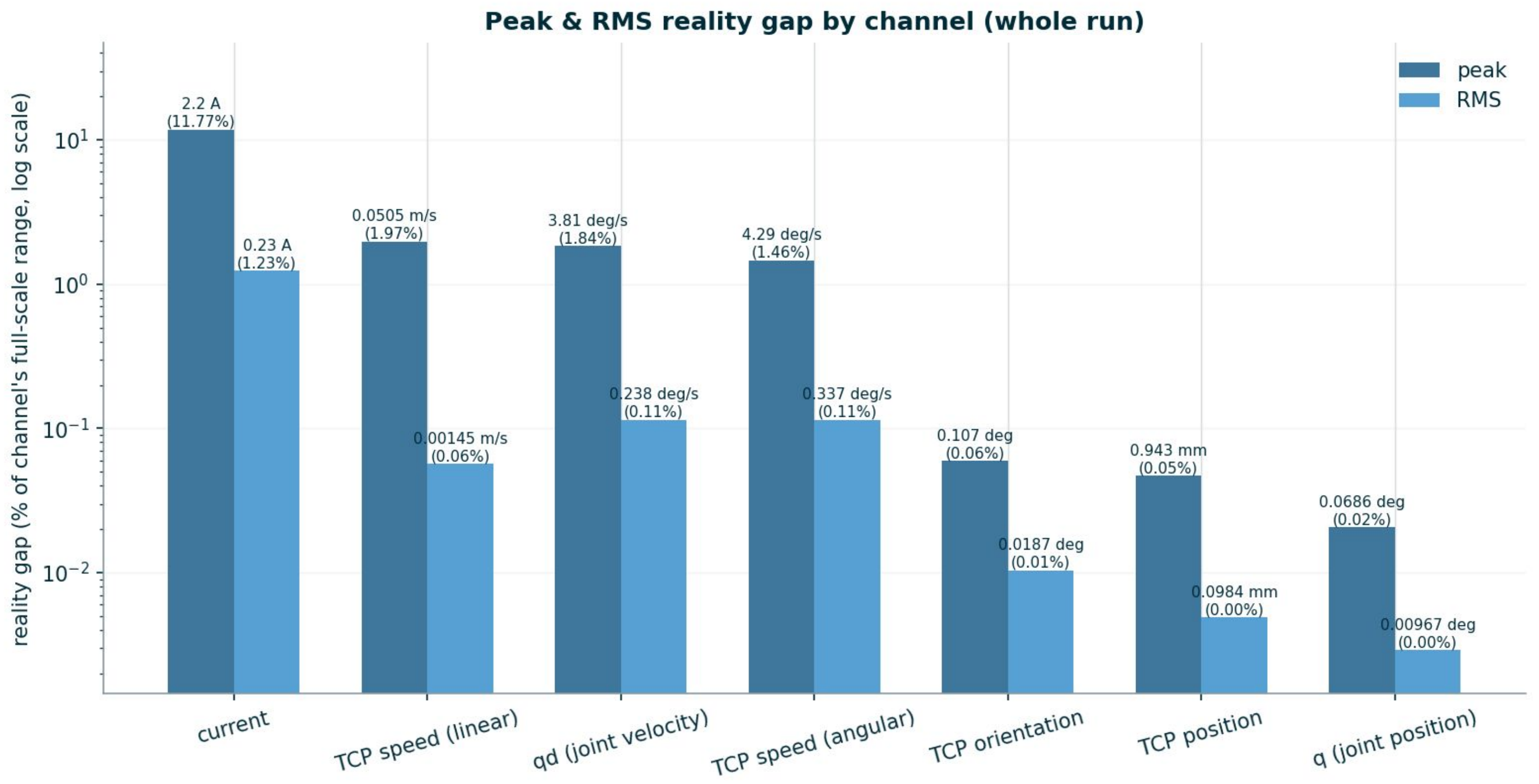
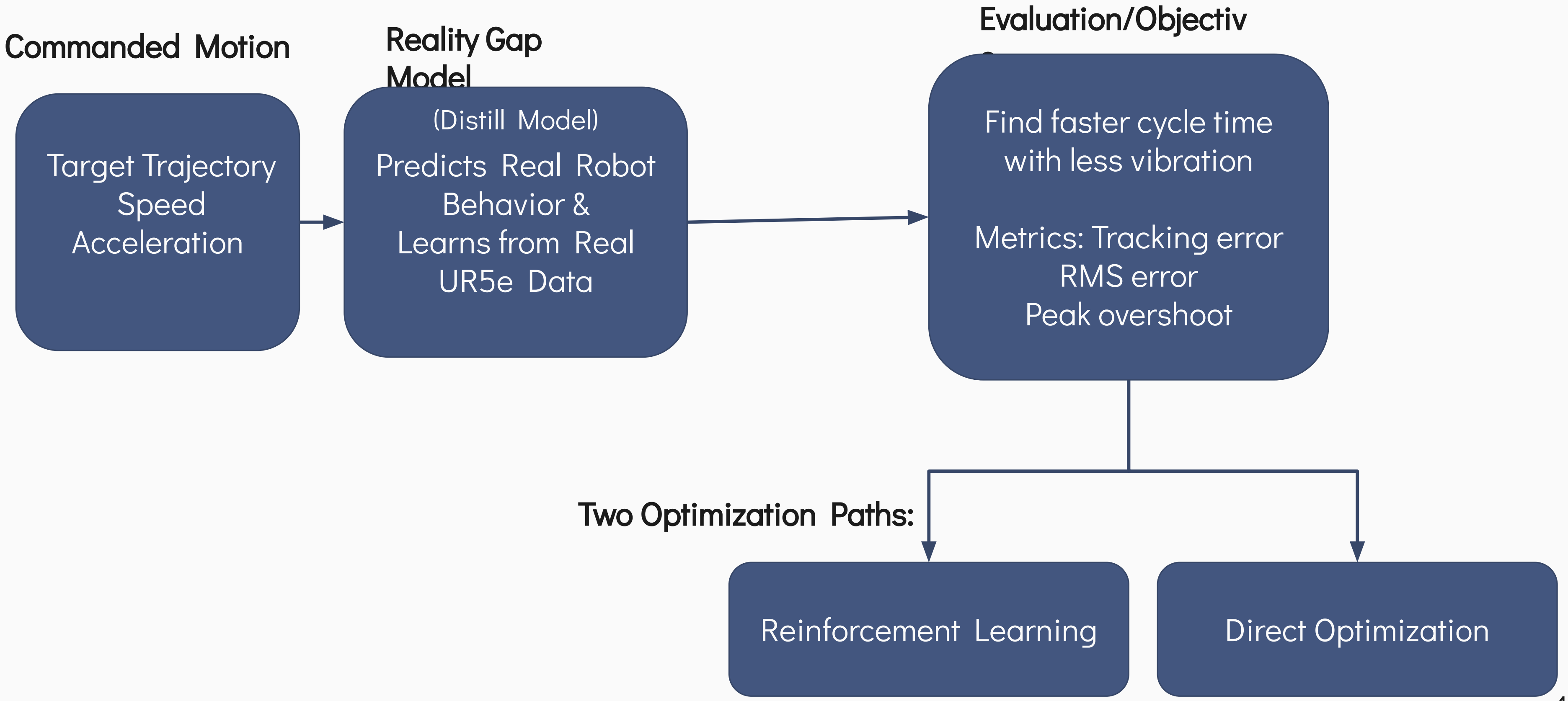
The Reality Gap



The Reality Gap

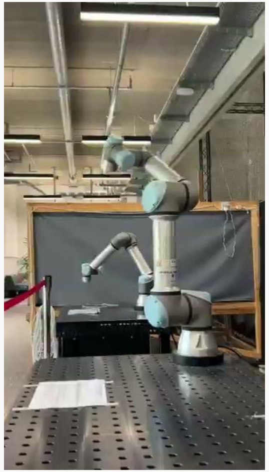


Optimization Pipeline



Collecting Data

- **48 URScript motion files total**
- 15 base / handwritten motion scripts
- 33 generated slow / medium / fast variants
- Most scripts recorded twice



Motion Set	What Varies	Why
Workspace / reach motions	TCP position, joint posture, near-body vs extended-arm poses	Capture workspace effects, gravity load, and leverage
Wrist motions	Wrist angles, tool orientation, small-joint motion	Capture wrist-specific overshoot and vibration
Short / combined moves	Move distance, speed, acceleration, multiple joints moving together	Capture start-stop settling, cycle-time tradeoffs, and joint coupling

Distill Model - RL

Datasets

- **Training Set:** 101 files covering a wide range of movement patterns and speeds.
- **Test Set (Held-out):** 12 unseen files used to evaluate the model's accuracy.

Models Tested

- **Global Linear:** Linear model with shared coefficients across all joints.
- **Joint Linear:** 6 independent linear models (one for each joint).
- **Tree model:** 6 Gradient Boosted Trees for non-linear dynamics.

Model Inputs (Tree Model)

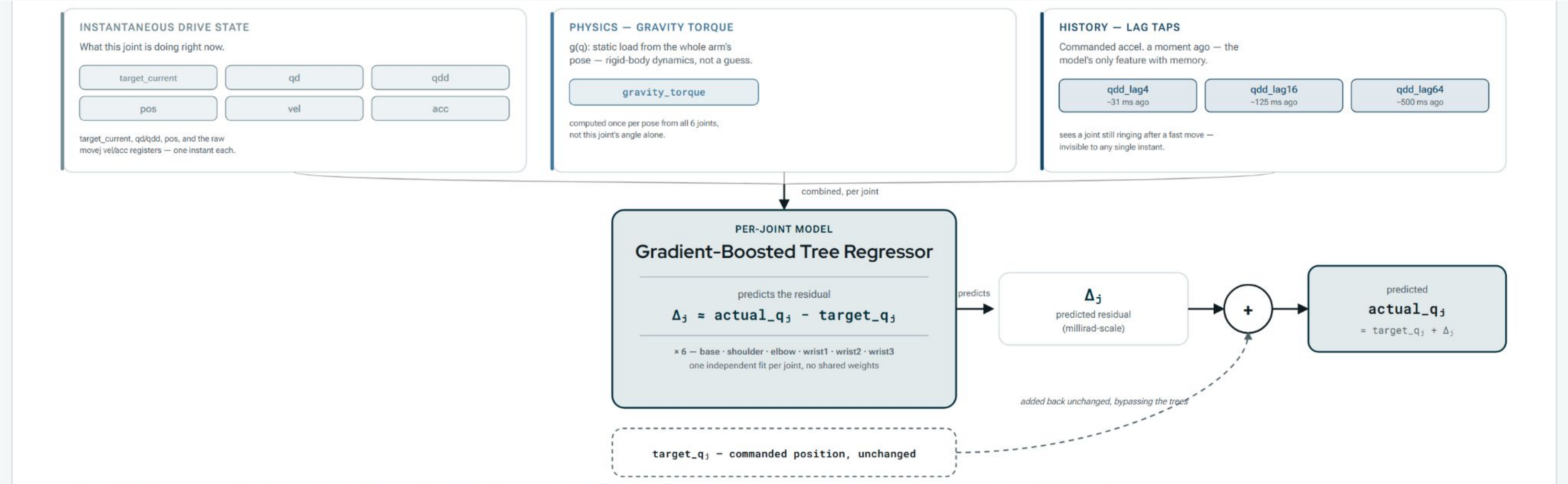
- **Target State:** Current, position, velocity, and acceleration values.
- **Torque Vector:** Torque needed at each joint to maintain current pose.
- **Lag History:** Commanded acceleration alongside history from earlier intervals (4, 16 & 64 samples).

Model Output

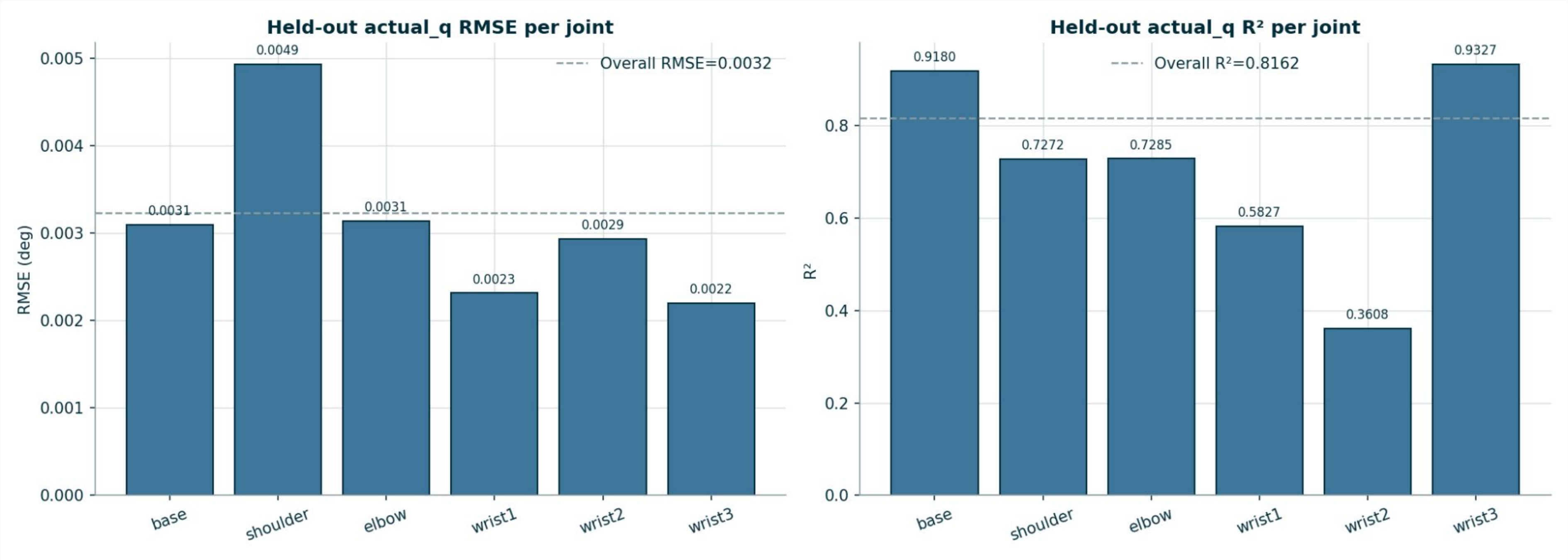
- **Position Residual:** Calculates the delta between actual and target positions.

Actual Position = Target Position + Residual

Distill Model Architecture – RL



Distill Model Results – RL



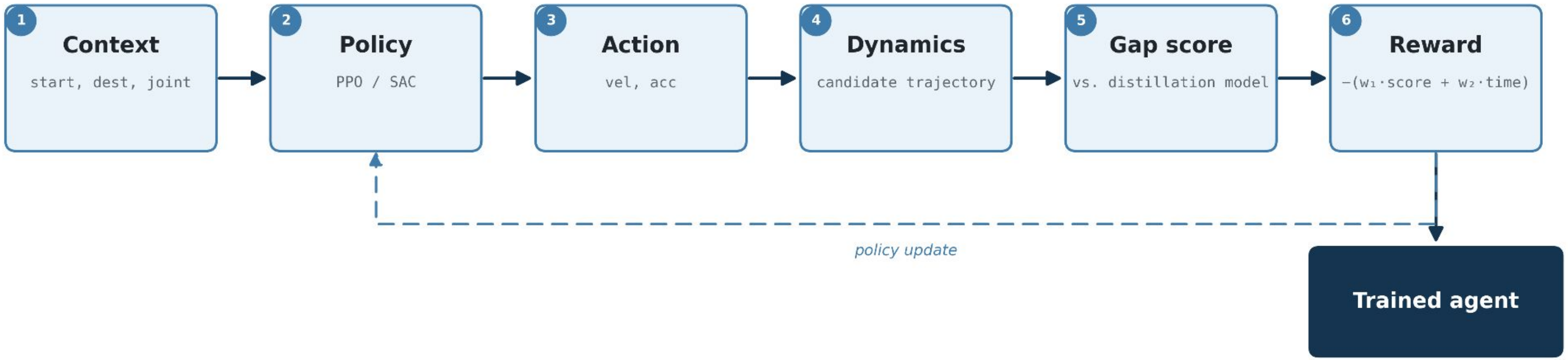
Path 1: Reinforcement Learning

- Goal: Minimize Cost

$$\mathbf{Cost} = 40000 * \mathbf{score} + \mathbf{cycle\ time}$$

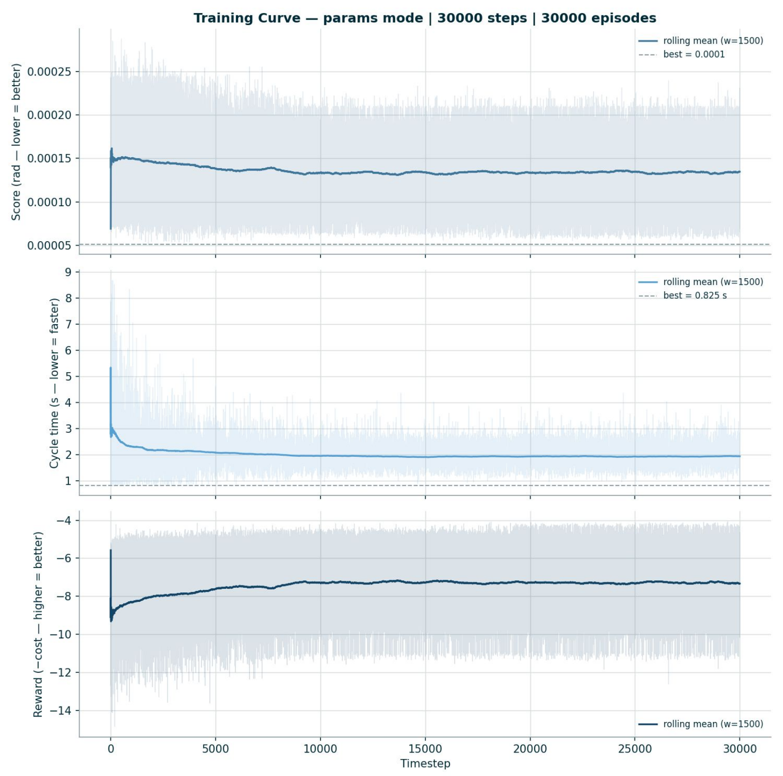
- Score: RMS error in joint position over the settling window (radians)
- Cycle time: Total time needed to complete a motion (seconds)
- Algorithms tested: SAC & PPO
- 30000 steps
- Search space: Position & Acceleration (provided to *movej* command)
- Trained on 3 movement patterns
  - Horizontal swing
  - Shoulder swing
  - Vertical swing
- Tested against a triangular baseline motion

RL Training Loop

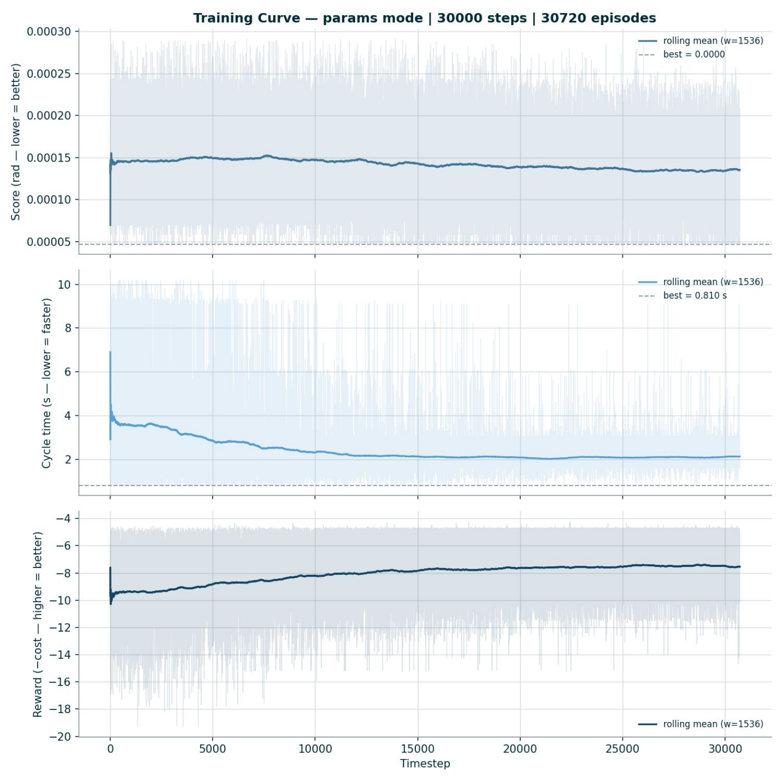


RL Analysis

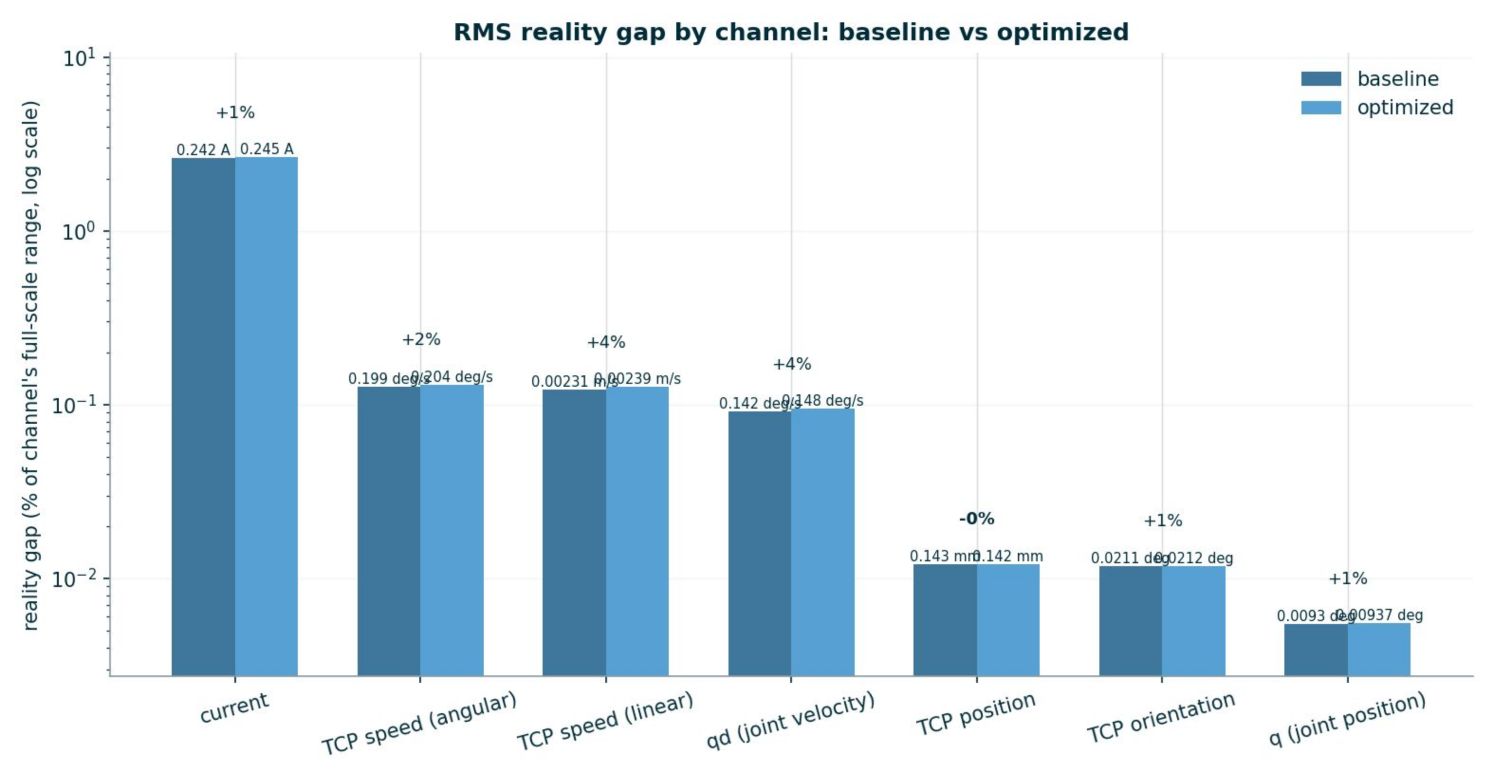
SAC



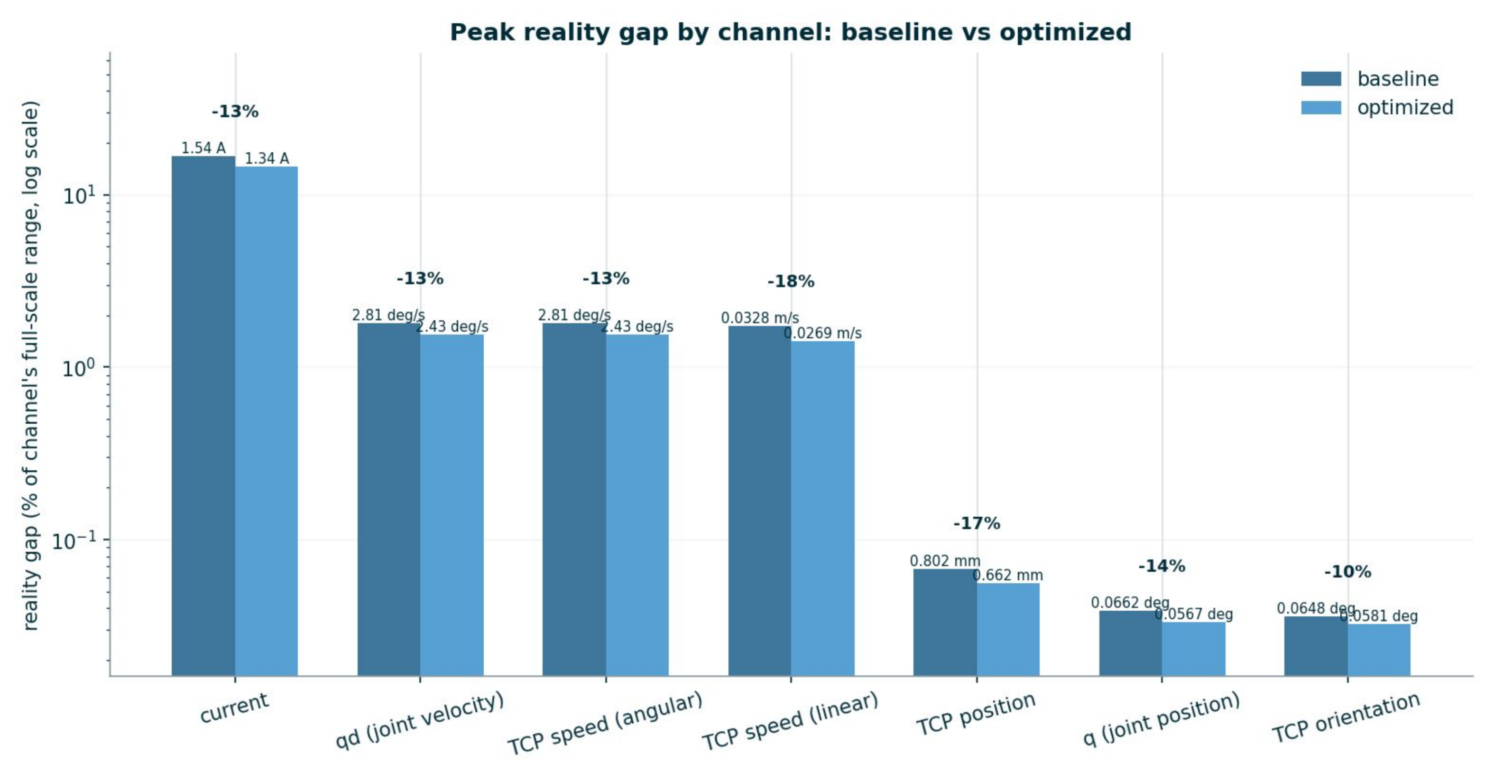
PPO



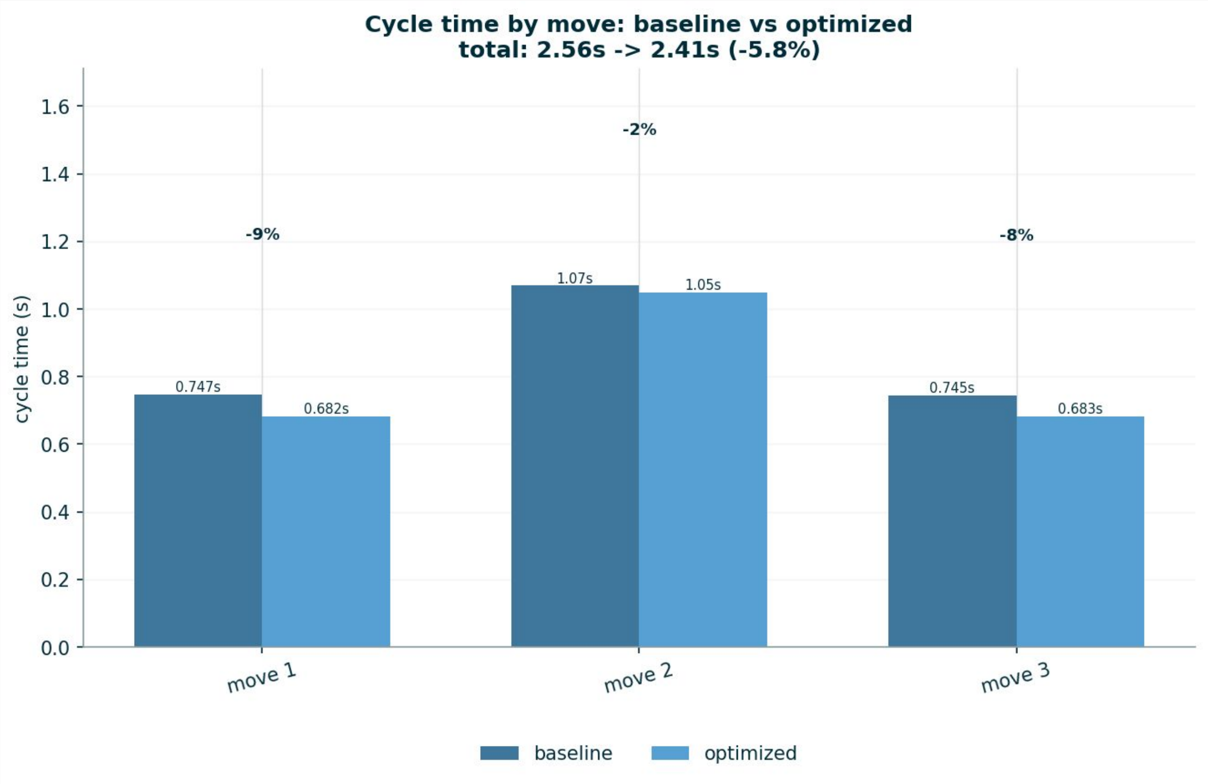
RL Results – SAC



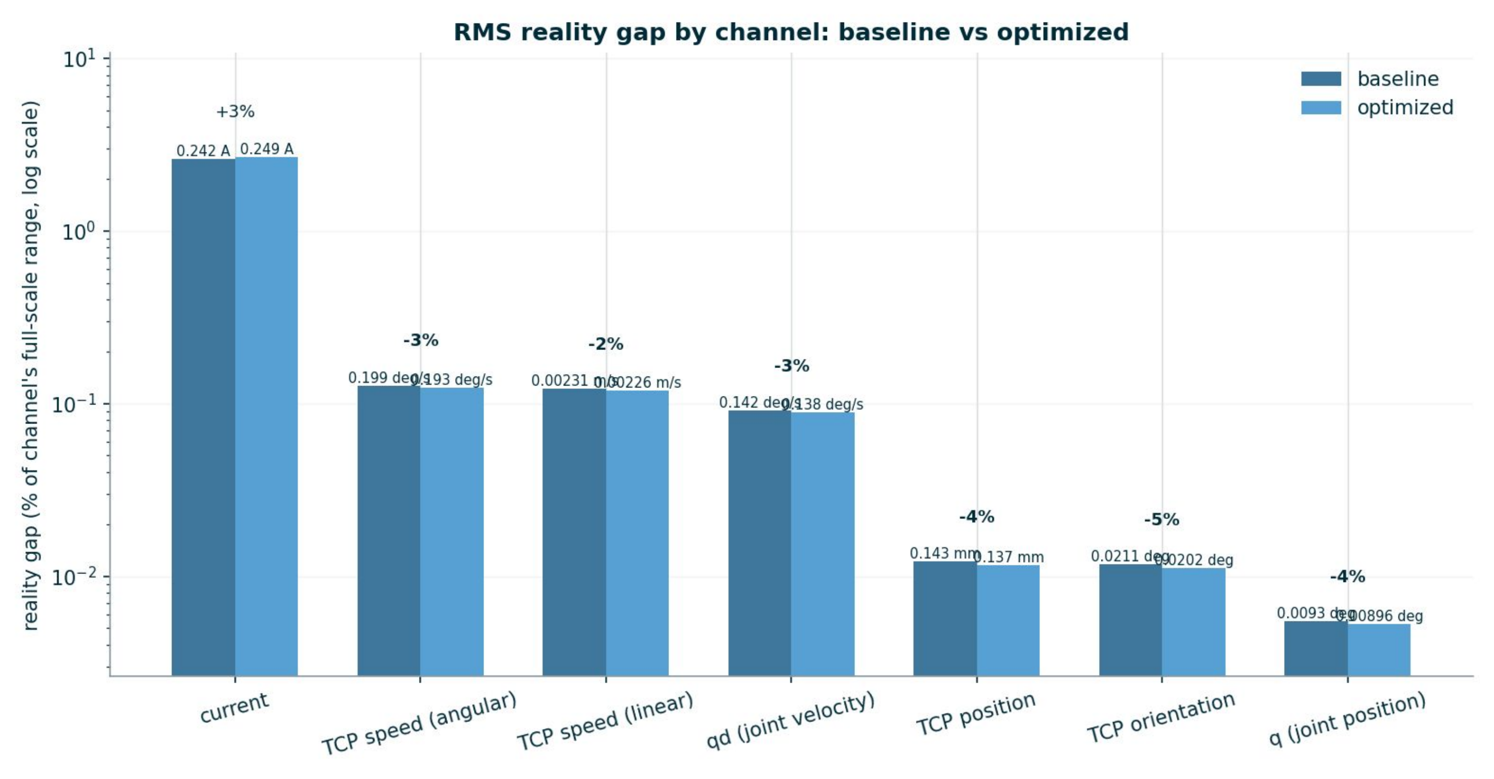
RL Results – SAC



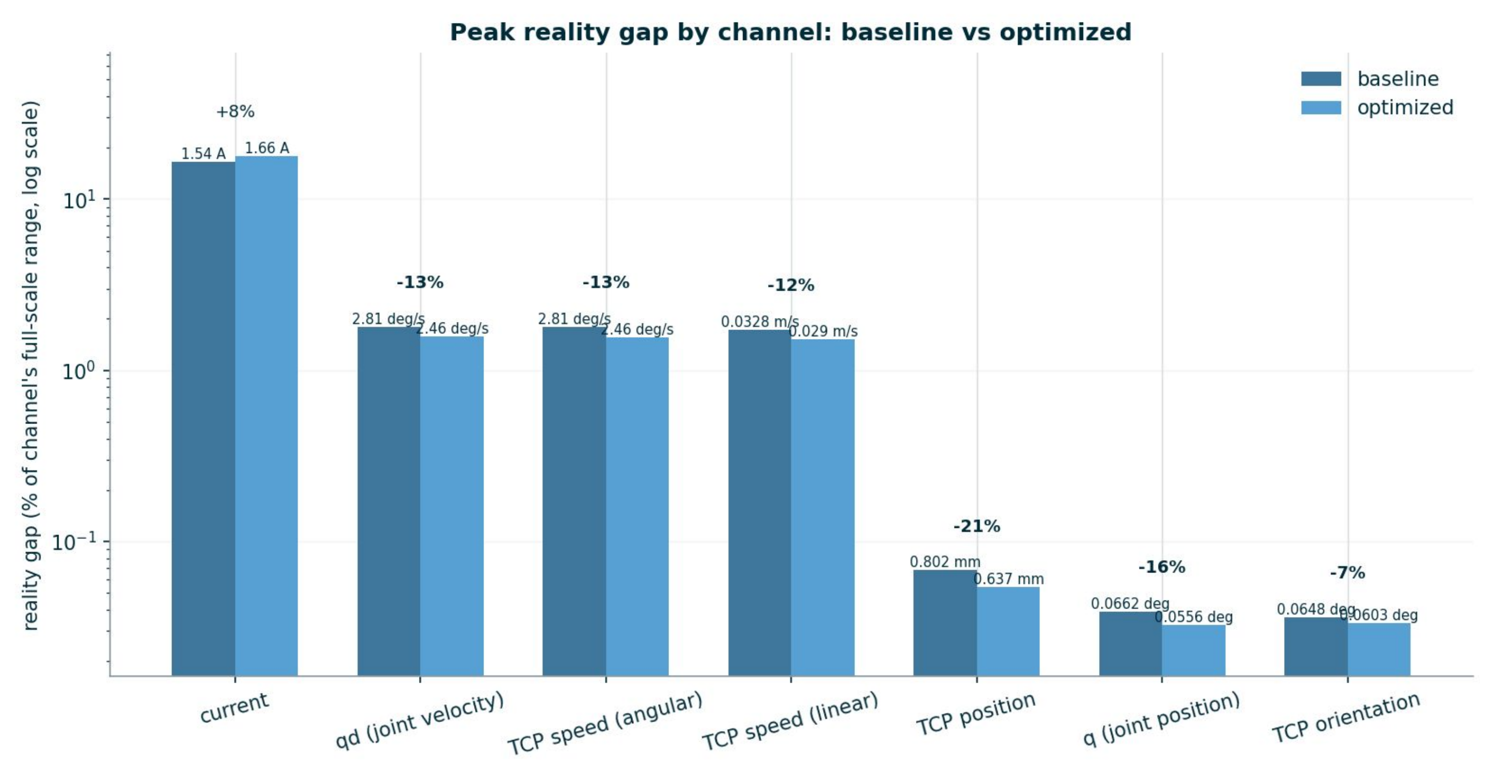
RL Results – SAC



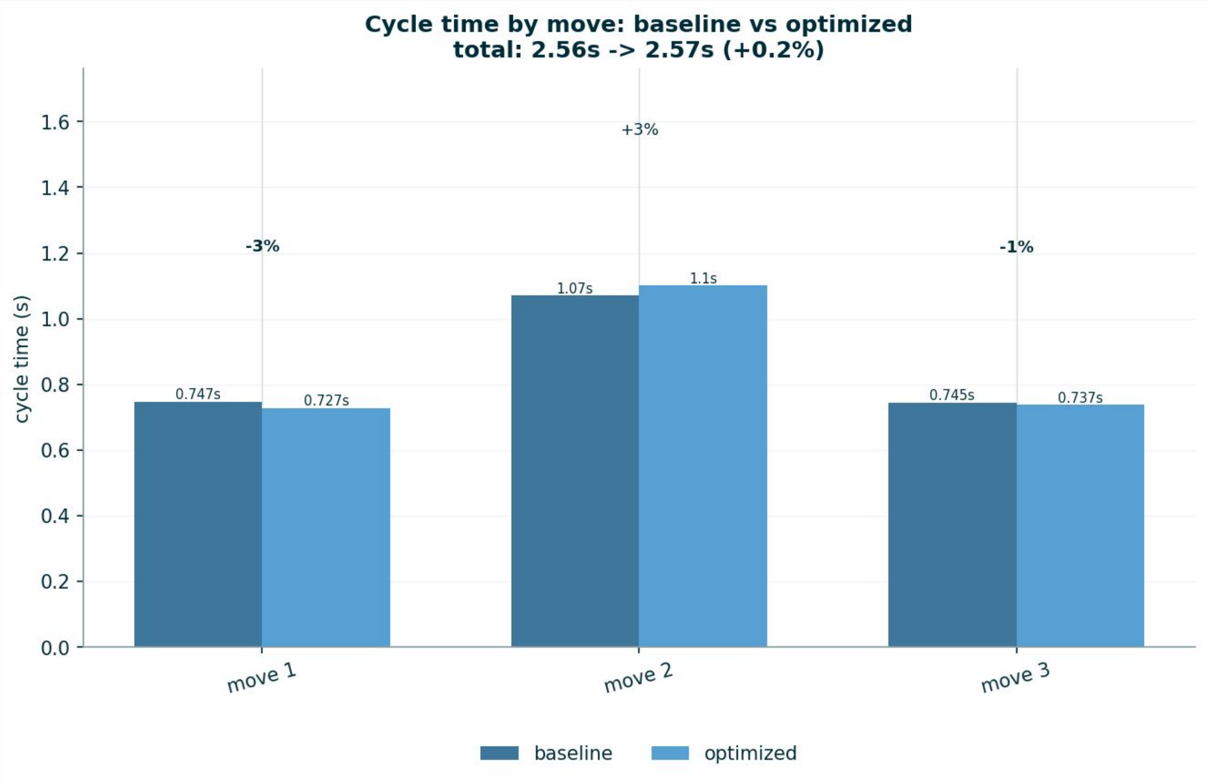
RL Results – PPO



RL Results – PPO



RL Results – PPO



RL Results Overview

- SAC
  - Increases RMS error (1 - 4 %)
  - Reduces peak position error (10 - 18%)
  - Reduces cycle time by 5.8%
- PPO
  - Reduces RMS error (2 - 5 %)
  - Reduces peak position error (7 - 21%)
  - Increases cycle time by 0.2%
- Final choice
  - PPO if we are more concerned about accuracy
  - SAC if we are more concerned about speed

Distill Model – Direct Optimization

Inputs
sin q, cos q, qd, qdd, qddd

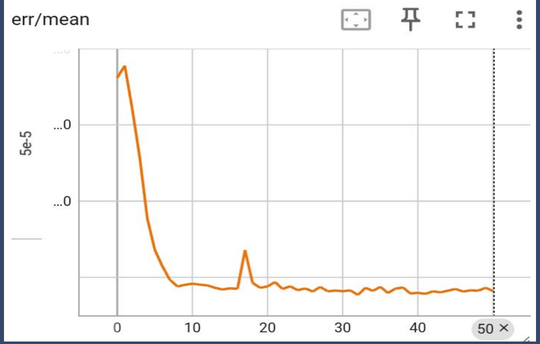
Distill Model
causal dilated 1D Convolutions
joints + features are channels

Output
12 channels = 6 gap + 6 var

Loss
First epochs: MSE(gap_error)
Later epochs: Negative log-likelihood
$\propto$ ½ log(var) + (gap_error)² / (2·var)

Uncertainty
total_var = aleatoric_var (model output) + epistemic_var (ensemble_mismatch)

Model Performance
Good on train data, e.g., 0.304 vs 0.306 mrad RMSE



Path 2: Direct Optimization

loss.backward() + opt.step()
to optimize trajectory

Trajectory parameterization
timesteps or B-Spline trajectory offset

Distill Model
predicts joint position reality gap + uncertainty

Loss
cycle time + k * gap + penalties

gap_rsme = RMSE (gap + uncertainty)

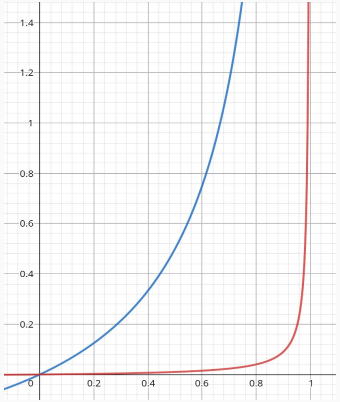
```
q0,q1,q2,q3,q4,q5,dt
0.000003,-1.399998,1.399998,-1.570000,-1.570000,0.000000,0.008000
0.000023,-1.399986,1.399986,-1.570000,-1.570000,0.000000,0.008000
0.000079,-1.399953,1.399953,-1.570000,-1.570000,0.000000,0.008000
0.000188,-1.399887,1.399887,-1.570000,-1.570000,0.000000,0.008000
0.000368,-1.399779,1.399779,-1.570000,-1.570000,0.000000,0.008000
```

Initialized by converting .script to .path with URSim

```
movej(CORNER_1, a=acc, v=vel)
sleep(0.8)
movej(CORNER_2, a=acc, v=vel)
sleep(0.8)
```

penalties = barrier(joint_velocities) + barrier(joint_accelerations) + barrier(tool_speed)

barrier gets sharper during optimization



Direct Optimization – Results – Video

Direct Optimization – Results – Statistics

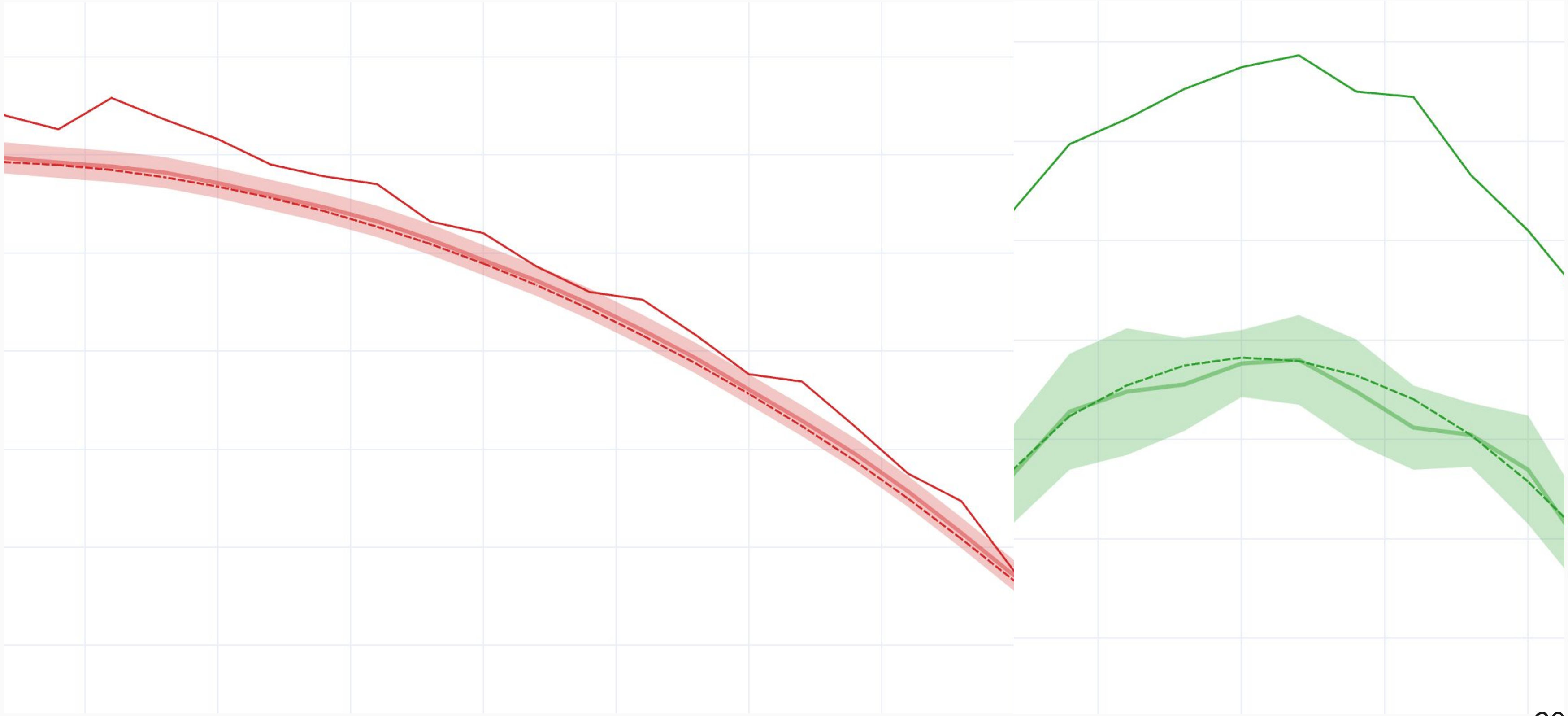
triangle

run	gap [mrad]	predicted_gap [mrad]	cycle [s]	loss
triangle.stream.csv	0.1288	0.0492	6.616	13.2320
triangle.retime.stream.csv	0.1503	0.0504	3.928	10.7029

shoulder_swing

run	gap [mrad]	predicted_gap [mrad]	cycle [s]	loss
shoulder_swing.stream.csv	0.1826	0.0562	3.168	6.3360
shoulder_swing.retime.stream.csv	0.1838	0.0602	2.864	6.2555

Direct Optimization – Results – Insights



Learnings – Challenges – Future Work

1. Bad training data (to small gaps), data distribution shift (faster trajectories, .script to .path, etc.)
   → must align better with inference distribution
   → exploration phase, retrain/finetune, exploitation phase
2. Position gap is noisy and often negligible
   → learning uncertainty is key, focus on large gaps
3. Tune architecture, hyperparameters, and other details (optimizer, barrier function, servoj settings, etc.)
4. Pareto front instead of trade-off factor